

Aishwarya Hareesh Rao

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Embedded Systems · Firmware & RTOS · TinyML · Edge AI · Real-Time Signal Processing · VLSI Design

EDUCATION

Arizona State University

M.S. Electrical Engineering

Tempe, Arizona · Jan 2025 – Dec 2026

Visvesvaraya Technological University

B.E. Electronics & Communication Engineering

Belagavi, India · Aug 2020 – Aug 2024

SKILLS

Languages: Python, C, C++, Embedded C, MATLAB, Verilog, VHDL, System Verilog

AI / ML & TinyML: TensorFlow, TFLite Micro, PyTorch, scikit-learn, Model Quantization, CNNs, Transfer Learning

Computer Vision & Signal Processing: OpenCV, dlib, Facial Landmark Detection, Real-Time Video Processing, Signal Processing, Time-Series Analysis

Embedded & Robotics: Arduino, ARM Cortex-M4, ROS2, Gazebo, rviz, Sensor Fusion, LiDAR, RGB-D Camera, Point Cloud Processing, PCB Design & Fabrication

VLSI & EDA: Cadence Virtuoso, HSPICE, StarRC, DRC/LVS Verification, Full-Custom Layout, Post-Layout Timing Analysis, Parasitic Extraction, Heterogeneous Integration, Semiconductor Packaging

Tools: Git, Linux, Jupyter, VS Code, ANSYS, Anomaly Detection, Closed-Loop Control

EXPERIENCE

Center for Artificial Intelligence and Robotics (DRDO)

Bangalore, India

Project Intern

Sep 2023 – Nov 2023

- Developed a **multi-sensor fusion pipeline** for **autonomous ground vehicle** object detection by integrating **LiDAR (360° FoV, 0.2–100 m range)**, **RGB-D camera (0.1–30 m depth)**, and Radar data streams using ROS2 and rviz for real-time 3D point **cloud visualization**.
- Built and validated **custom UGV simulation environments** in **Gazebo** with 3D robot models designed in **Blender**, enabling repeatable **autonomous navigation testing** across cluttered obstacle scenarios without physical hardware dependency.
- Implemented **ROS2 sensor data synchronization** across 3 modalities (LiDAR, camera, radar) using **tf2 transforms** and topic-level message filtering, enabling **time-aligned fusion** for reliable **obstacle detection** and **environmental mapping** at **30 FPS** camera input.

PROJECTS

IMU-Based Posture Detection System

Aug – Dec 2025 · ASU

- Achieved **99.99% classification accuracy** across 5 posture classes by training feedforward and **1D-CNN** models on **9-axis** IMU data with z-score normalization, Gaussian augmentation, and sliding-window feature extraction (100 samples, 50% overlap).
- Deployed **3 sensor-specific models (~6K parameters, 19–65 KB)** to an **ARM Cortex-M4** via TFLite Micro, delivering **sub-150ms on-device inference with no cloud dependency** within a **32 KB** arena through an automated Keras-to-C-header conversion pipeline.

Integrate-and-Fire Neuron VLSI Design

Jan – May 2025 · ASU

- Designed and verified a **digital integrate-and-fire neuron** from gate-level schematic to physical layout in **391 μm^2** , achieving a **0.5 ns minimum clock period** and **5.79 ps worst-case propagation delay** as validated through HSPICE transient simulation and StarRC parasitic extraction.
- Delivered **DRC-clean, LVS-passing layouts** for the IF neuron and D flip-flop by applying transistor-pair matching, signal shielding, and parasitic-aware routing in Cadence Virtuoso, meeting all timing constraints across pre-layout and post-layout verification.

Driver Distraction & Drowsiness Detection System

Jan – Aug 2024 · RVITM

- Achieved **86% detection accuracy** across **1,600 observations** by building a **real-time computer vision** pipeline that extracted **Eye Aspect Ratio (EAR)**, **PERCLOS**, **gaze ratio** (pupil-to-corner displacement), and **3-axis head pose estimation** (yaw, pitch, roll) from **30 FPS** dashcam video using **facial landmark detection**.
- Reduced driver alert response time to **under 3 seconds** by building a **closed-loop control system** that fused drowsiness signals (**EAR < 0.20 for $\geq 2.5\text{s}$**) with gaze deviation tracking to trigger buzzer actuation and automated SMS notifications, with no cloud dependency.